FlowPilot Supporting Information Figure and Content Plan

Prepared after review of main_manuscript.pdf and the current FlowPilot codebase
Version date: 2026-08-11 | Proposed scope: 38 figures, 14 tables, 12 text sections

Editorial decision captured in this plan

Revised main Figure 5 will retain the photoredox Giese/aerobic-oxidation story and add wet-lab refinement data. Current main Figure 6 (alpha-bromination with inline thiosulfate quench) moves to the ESI. Revised main Figure 6 will present the two replacement DPDTC protocols with wet-lab results. The ESI preserves full protocols, calculations, logs, unsuccessful cycles, ablation evidence, and analytical details.

Executive assessment

The current manuscript tells a clear six-figure story: architecture (Figure 1), corpus/rules (Figure 2), retrieval (Figure 3), model/council behavior (Figure 4), photoredox case study (Figure 5), and bromination case study (Figure 6). The revised paper should keep the main text compact and shift detailed validation into a deliberately ordered ESI. The ESI should not merely collect unused plots; each section below supports a specific claim in the main manuscript.

Main-manuscript consequences

Location	Required revision
Abstract	Replace the claim that the two end-to-end case studies are Giese oxidation and alpha-bromination. State the final wet-lab case set and distinguish literature comparison from experimental validation.
Introduction contributions	Update the named validation cases and add matched architecture ablation, standardized intake, inventory-bound design, and closed-loop refinement only if these are claimed in Results.
Figure 5 Results	Add wet-lab cycles and measured outcomes. Move full protocol, hardware, gas-basis arithmetic, unsuccessful cycles, and analytical data to Figures S34-S35 and Tables S12-S14.
Figure 6 Results	Replace bromination with the two DPDTC protocols plus wet-lab data. Move the current bromination text and figure to ESI Figures S32-S33.
Discussion	Remove bromination-specific evidence from the main narrative or cite ESI. Add what experimental feedback changed and clearly separate topology success from yield prediction.
Limitations	Delete the statement that FlowPilot does not execute experiments once wet-lab validation is included. Retain that experiments are human-executed and that the system proposes screens. Update the obsolete statement that gas holdup is not modeled if the current implementation is reported.
Intensification claims	The current code is evidence-first and no longer forces intensification without an explicit objective. Revise language claiming that every design is systematically forced toward shorter residence time.
Conclusion	Replace the original two-case conclusion with the final

	experimental cases and report failed/iterative screens, not only the best yields.
Methods	Add intake, inventory extraction/validation, final-design contract, gas-basis definitions, closed-loop update procedure, ablation endpoints, reproducibility, and wet-lab methods.
Model names and dates	Use exact deployed model identifiers consistently. Keep architecture-effect and model-effect claims separate.

Recommended ESI structure

- S1. Workflow, data contracts, and reproducibility: Document what enters the system, which component owns each decision, and how a complete run can be reconstructed from stored artifacts.
- S2. Literature corpus, rule base, and retrieval: Provide the detailed distributions and retrieval controls underlying main Figures 2 and 3 without crowding the main manuscript.
- S3. Deterministic engineering, multiphase calculations, and inventory: Expose the equations and closure checks that distinguish FlowPilot from a prose-only LLM while documenting recent heat-transfer, gas-liquid, multistage, and inventory additions.
- S4. Council behavior, model matrix, and reproducibility: Provide the detailed evidence behind main Figure 4, including candidate trajectories, model dependence, repeated runs, and candidate-budget effects.
- S5. Matched ablation and architecture comparison: Place the complete ablation in the ESI using the defensible matched-model design: the same model, protocol, objective, inventory, condition, and repeat input for one-shot and full FlowPilot.
- S6. Case studies and wet-lab validation: Preserve the displaced bromination case and provide the complete protocols, design histories, inventory constraints, experimental conditions, and raw analytical evidence behind revised main Figures 5 and 6.

Proposed supporting figures

Numbering principle. Figures proceed in the same logical order as the manuscript: system inputs and provenance, knowledge grounding, deterministic engineering, council behavior, ablation, and finally chemical/experimental validation. Existing assets are retained only when their benchmark design and labels remain compatible with the final Methods.

S1. Workflow, data contracts, and reproducibility

Document what enters the system, which component owns each decision, and how a complete run can be reconstructed from stored artifacts.

Figure S1. Extended FlowPilot architecture and authority boundaries

Purpose	Expand main Figure 1 into the complete executable architecture, including standardized intake, chemistry interpretation, retrieval, deterministic calculations, council, inventory reconciliation, final validation, topology rendering, and autosave.
Recommended panels	(a) end-to-end modules; (b) data objects passed between modules; (c) authority hierarchy: measured evidence > hard constraints > protocol facts > hypotheses > model inference; (d) executable versus blocked output paths.
Code/data source	Existing starting asset: ablation_results/figures/23_full_flowpilot_pipeline.pdf. Update from flora_translate/main.py, schemas.py, final_design_contract.py, and topology_compiler.py.
Production status	Regenerate

Figure S2. Component-level schema and data-flow contracts

Purpose	Show that the LLM agents exchange typed objects rather than unconstrained prose and identify the authoritative source for each final field.
Recommended panels	BatchRecord -> ChemistryPlan -> FlowProposal -> DesignCalculations -> council candidate -> inventory-bound topology -> FinalDesignContract.
Code/data source	Generate from flora_translate/schemas.py and representative full_result.json artifacts.
Production status	New from code

Figure S3. Reproducible standardized intake workflow

Purpose	Explain the conversational intake LLM while demonstrating fixed question identifiers and deterministic readiness rules.
Recommended panels	(a) protocol submission; (b) fixed IDs Q-BATCH-001 through Q-PREF-001; (c) answered/unavailable states; (d) frozen DesignInputPackage; (e) design gate.
Code/data source	components/intake_wizard.py, flora_translate/intake_agent.py, and intake tests.
Production status	New from code

Figure S4. Representative frozen DesignInputPackage and evidence hierarchy

Purpose	Provide a readable example of how protocol facts, objectives, historical measurements, inventory constraints, hypotheses, and safety limits are stored before design.

Recommended panels	Annotated JSON excerpt plus a provenance map showing which package sections influence upstream chemistry, deterministic calculations, and council review.
Code/data source	Use a redacted intake_package.json from outputs/benchmarks/khu_three_protocols_canonical_20260810_161346.
Production status	Existing data; compose

Figure S5. GUI workflow and result provenance

Purpose	Document the user-visible path from intake and inventory selection to final tabs while demonstrating that every tab reads the same FinalDesignContract.
Recommended panels	Intake, inventory selection, final summary, engineering design, stream assignment, council trace, process topology, and raw JSON.
Code/data source	Capture with Streamlit/Playwright from pages/flora_design_unified.py and components/*.py after the final GUI regression.
Production status	New screenshot set

Figure S6. Autosave artifact tree and run-level provenance

Purpose	Demonstrate reproducibility through exact prompts, model events, inventory profile, raw result, canonical result, topology files, validation audit, logs, and checksums.
Recommended panels	(a) folder tree; (b) artifact-to-claim mapping; (c) checksum verification; (d) model/component event count.
Code/data source	outputs/gui_runs, outputs/benchmarks/khu_three_protocols_canonical_20260810_161346, and flora_translate/gui_autosave.py.
Production status	New from existing artifacts

S2. Literature corpus, rule base, and retrieval

Provide the detailed distributions and retrieval controls underlying main Figures 2 and 3 without crowding the main manuscript.

Figure S7. Full literature-corpus composition

Purpose	Expand the corpus overview beyond the selected categories shown in main Figure 2.
Recommended panels	Reaction class, reactor type, tubing material, phase regime, number of streams, temperature, pressure, residence time, publication year, and source-paper counts.
Code/data source	Regenerate from the curated literature records used for the 464-paper corpus.
Production status	Regenerate

Figure S8. Engineering rule-base coverage and severity structure

Purpose	Document all 2,537 rules, their categories, severity levels, source type, and chemistry-class associations.
Recommended panels	Category counts; severity distribution; category x chemistry

	heatmap; rule-source distribution; sparse-domain map.
Code/data source	flora_fundamentals knowledge store and rule export used for main Figure 2.
Production status	Regenerate

Figure S9. Plan-aware retrieval algorithm and tier transitions

Purpose	Show the complete hard-filter, relaxed-filter, and semantic-fallback logic with field-aware reranking.
Recommended panels	Algorithm diagram, query fields, fallback triggers, weighted score, and one worked query.
Code/data source	Retrieval implementation plus main Figure 3 source data.
Production status	New detailed schematic

Figure S10. Retrieval performance, reranking shifts, and leakage controls

Purpose	Support the retrieval claim with family-level results, rank changes, and proof that hidden benchmark references are excluded.
Recommended panels	Top-k family matching; rank-shift distribution; field-score contribution; similarity-to-reference heatmap; per-case hidden-source exclusion audit.
Code/data source	benchmark manuscript_comparison figures and ablation_test/tests/test_retrieval_exclusion.py.
Production status	Existing analyses; combine

S3. Deterministic engineering, multiphase calculations, and inventory

Expose the equations and closure checks that distinguish FlowPilot from a prose-only LLM while documenting recent heat-transfer, gas-liquid, multistage, and inventory additions.

Figure S11. Nine-step deterministic engineering calculator

Purpose	Provide a detailed version of the calculator sequence and identify the inputs, equations, outputs, and failure gates at every step.
Recommended panels	Kinetics/residence time; reactor sizing; fluid dynamics; pressure drop; mixing/mass transfer; heat transfer; vapour pressure/BPR; process metrics; final validation.
Code/data source	flora_translate/design_calculator.py and components/design_steps.py.
Production status	New from code

Figure S12. Engineering arithmetic closure and unit-consistency validation

Purpose	Show deterministic tests for V = Q tau, coil geometry, pressure drop, Re, Pe, heat-transfer area, and stage-total reconciliation.
Recommended panels	Expected-versus-computed parity plots and pass-rate bars across tests/benchmark outputs.
Code/data source	flora_translate/tests and canonical case audit.json files.
Production status	New from tests

Figure S13. Heat-transfer calculation workflow and sensitivity

Purpose	Document the recent UA, area-to-volume, heat-generation/removal, and thermal-Damkohler calculations.
Recommended panels	Equation schematic; UA versus tubing ID/volume; heat-removal margin; representative thermal and photochemical cases; warning thresholds.
Code/data source	Design calculation heat_transfer_metrics and associated tests.
Production status	New from code

Figure S14. Gas-liquid bookkeeping: inlet/STP versus in-channel basis

Purpose	Make the gas-flow convention unambiguous and show why pressure-corrected gas flow changes channel residence time but not inlet equivalents.
Recommended panels	STP feed; pressure/temperature correction; inlet residence; channel residence; O2-equivalent closure; example from the photoredox oxidation stage.
Code/data source	Existing ablation_results/figures/15_gas_bookkeeping.pdf plus current multiphase code and tests.
Production status	Update existing

Figure S15. Multistage residence-time and stream-flow accounting

Purpose	Show how upstream effluent and newly added feeds are propagated without double counting and how per-stage residence times sum to the final total.
Recommended panels	Two-stage liquid example; gas-addition example; quench-only operation; serial-reactor example; stage arithmetic audit.
Code/data source	flora_translate/multistage_inventory.py, topology compiler, and three-protocol canonical audits.
Production status	New from code

Figure S16. Inventory ingestion and canonical profile generation

Purpose	Explain how PDF/PPT/DOCX/free-text equipment descriptions become a validated, reusable LabInventory JSON profile.
Recommended panels	Document upload; LLM extraction; null normalization; Pydantic validation; human confirmation; saved profile; import into design.
Code/data source	Inventory Manager GUI and deliverables/flowpilot_inventory_management_20260804.
Production status	Existing schematic; adapt

Figure S17. Inventory capability coverage and deterministic allocation

Purpose	Demonstrate selection among pumps, tubing, reactors, lights, MFCs, mixers, BPRs, connectors, and temperature limits, including blocked and assumed-accessory states.
Recommended panels	Capability matrix; allocation graph; accepted design; rejected unavailable reactor; temperature clamp; unresolved topology; final instrument manifest.
Code/data source	khu_inventory_updated_flowpilot.json, ablation_results/figures/12_inventory_feasibility.pdf, and topology

	artifacts.
Production status	Regenerate with final inventory logic

S4. Council behavior, model matrix, and reproducibility

Provide the detailed evidence behind main Figure 4, including candidate trajectories, model dependence, repeated runs, and candidate-budget effects.

Figure S18. Council candidate lifecycle and bounded revision

Purpose	Show generation, domain scoring, deterministic rejection, specialist revision, skeptic audit, selection, and final recomputation for one complete run.
Recommended panels	Candidate Sankey/funnel; specialist decisions; rejected revisions; winning-candidate trajectory; final score components.
Code/data source	Council logs and llm_events.jsonl from a representative model-matrix run.
Production status	New from logs

Figure S19. Complete 4 x 4 model-matrix engineering outcomes

Purpose	Move the detailed per-cell metrics supporting main Figure 4 into the ESI.
Recommended panels	Residence time, flow rate, reactor volume, tubing ID, BPR, validation status, screen-required status, and concern count heatmaps.
Code/data source	benchmark/data/model_matrix_benchmark_20260504_101732/ visualizations.
Production status	Existing; assemble

Figure S20. Pre/post-council repeatability and metric covariance

Purpose	Show all repeats behind the representative radar plot and quantify run-to-run variation rather than displaying only one aggregate polygon.
Recommended panels	Repeat overlays; paired deltas; coefficient-of-variation heatmap; metric-correlation matrix; design-regime scatter.
Code/data source	prepost_radar outputs and ablation_results/figures/17_metric_correlation.pdf.
Production status	Existing analyses; combine

Figure S21. Candidate-budget effects across B = 1, 6, 12, and 24

Purpose	Provide the complete distributions behind main Figure 4e-f and document outlier/pathology rates.
Recommended panels	Design-family counts; metric distributions; revision/disqualification activity; CV heatmap; pathology rate; completion rate.
Code/data source	benchmark/data/ protocol_budget_benchmark_20260427_181556/visualizations.
Production status	Existing; assemble

Figure S22. Runtime, token use, LLM calls, and stage-level cost

Purpose	Separate computational cost from design quality and identify which stages dominate execution.
Recommended panels	Runtime; tokens; calls; stage breakdown; cost-quality scatter; failed/retried requests.
Code/data source	Model-matrix visualizations and ablation_results/figures/09, 10, 16, and 19.
Production status	Existing; update model names

S5. Matched ablation and architecture comparison

Place the complete ablation in the ESI using the defensible matched-model design: the same model, protocol, objective, inventory, condition, and repeat input for one-shot and full FlowPilot.

Figure S23. Ablation study design and frozen endpoint calculation

Purpose	Define the matched comparisons, feasible/infeasible inventories, repeats, primary executability endpoint, secondary quality dimensions, and interpretation boundaries.
Recommended panels	Study matrix; input matching; endpoint decision tree; score calculation; frozen-weight statement.
Code/data source	ablation_results/presentation/ flowpilot_vs_oneshot_qwen_gpt_20260810/figures/ 03_score_calculation.pdf and benchmark manifests.
Production status	Existing; adapt for manuscript

Figure S24. Qwen 27B one-shot versus Qwen 27B within FlowPilot

Purpose	Isolate the architecture effect while holding the local base model fixed.
Recommended panels	Executable rate; correct-block rate; win/tie/loss; paired case scores; repeat dispersion.
Code/data source	qwen_gpt_architecture_comparison_v2 and presentation figures.
Production status	Existing publication candidate

Figure S25. GPT one-shot versus the same GPT model within FlowPilot

Purpose	Repeat the architecture comparison with the commercial model while preserving exactly matched inputs and scoring.
Recommended panels	Executable rate; correct-block rate; win/tie/loss; paired case scores; repeat dispersion.
Code/data source	qwen_gpt_architecture_comparison_v2 and presentation figures. Replace provider/model labels with the exact endpoint reported in Methods.
Production status	Existing publication candidate

Figure S26. Paired architecture uplift by chemistry family

Purpose	Show whether the architecture benefit is consistent across reaction families rather than driven by one protocol.
Recommended panels	Case-level paired deltas and chemistry-family cluster intervals for Qwen and GPT separately.

Code/data source	presentation/02_architecture_effects.pdf and comparison tables.
Production status	Existing publication candidate

Figure S27. Quality-dimension decomposition and schema-neutral sensitivity

Purpose	Address the criticism that FlowPilot wins only because it emits its own schema by showing both the frozen dimensions and a schema-neutral sensitivity analysis.
Recommended panels	Engineering closure; safety; inventory; evidence; assurance; actionability; formal validity; schema-neutral paired effect.
Code/data source	reduced_frontier_final_package figures 03, 11, and 12 plus presentation/04_quality_score_breakdown.pdf.
Production status	Existing; combine carefully

Figure S28. Hard-check and deployment-gate rates

Purpose	Keep scientific design quality separate from immediate deployment readiness and expose every failing gate.
Recommended panels	Geometry closure; gas bookkeeping; pump range; pressure; temperature; inventory assignment; topology; screen-required; deployment-ready rate.
Code/data source	reduced_frontier_final_package figures 06 and 07 and ablation_results/figures/29-30.
Production status	Existing; use current deterministic gates

Figure S29. Weight sensitivity, bootstrap uncertainty, and rank robustness

Purpose	Demonstrate which conclusions are robust to reasonable weighting choices and which remain value-dependent.
Recommended panels	Six prespecified weight schemes; 20,000-weight-set winner frequency; paired bootstrap intervals; per-case win/tie/loss.
Code/data source	presentation/05_weight_sensitivity.pdf and ablation_results/figures/27-28.
Production status	Existing; report as sensitivity, not proof

Figure S30. Evidence-first safeguards and correction of fallback dependence

Purpose	Show the before/after effect of removing forced intensification, preserving measured residence-time evidence, and deterministically validating final revisions.
Recommended panels	Council path before/after; residence-time evidence adherence; joint disposition/engineering success; safeguard-event counts.
Code/data source	stage3_evidence_first_validation_package figures 10-13.
Production status	Existing publication candidate

Figure S31. Agent call trace and proof of multi-agent execution

Purpose	Provide direct evidence that upstream, deterministic engineering, council specialists, skeptic, revision, and orchestrator components actually executed.
Recommended panels	Chronological call trace; component call counts; prompt/response hashes; representative bounded messages; retry/fallback annotation.

Code/data source	ablation_results/tables/agent_call_events.csv, reports/agent_trace_summary.md, and representative llm_events.jsonl.
Production status	Existing logs; compose

S6. Case studies and wet-lab validation

Preserve the displaced bromination case and provide the complete protocols, design histories, inventory constraints, experimental conditions, and raw analytical evidence behind revised main Figures 5 and 6.

Figure S32. Thermal alpha-bromination with inline thiosulfate quench

Purpose	Move the current main Figure 6 intact into the ESI so the original literature-comparison case remains part of the evidence record.
Recommended panels	Current panels: council deliberation, validated process flow diagram, and design-versus-literature comparison.
Code/data source	flent_fig_png/fig7.png and current main_manuscript.pdf Figure 6.
Production status	Existing; move intact

Figure S33. Bromination candidate space, mixing warning, and validation details

Purpose	Supply the quantitative background omitted from the moved composite figure and preserve the unresolved micromixing limitation.
Recommended panels	Candidate geometry map; mixing criteria; conversion estimates; selected-versus-rejected candidates; scale comparison; safety/quench topology audit.
Code/data source	Bromination benchmark result and council logs; multiphase validation protocol.
Production status	New from existing run

Figure S34. Photoredox Giese/aerobic oxidation: complete protocol and inventory-bound topology

Purpose	Provide the full chemistry, stage-separation logic, oxygen timing, no-inline-degasser constraint, instrument assignments, STP/in-channel gas flows, and residence-time arithmetic supporting revised main Figure 5.
Recommended panels	Batch protocol; chemistry mechanism/stream separation; initial topology; no-degasser inventory topology; instrument manifest; gas-basis calculation.
Code/data source	case_study1.png, scripts/run_case_study1_no_inline_degas.py, and canonical case-01 artifacts.
Production status	Existing data; regenerate after final experimental design is frozen

Figure S35. Photoredox case: full closed-loop experimental refinement history

Purpose	Expand the wet-lab panel added to main Figure 5 into a complete cycle-by-cycle record, including rejected below-pump-limit designs and inventory changes.
Recommended panels	Cycle timeline; reactor volume; liquid/O2 flow; inlet/channel residence; temperature/pressure; measured yield; agent feedback; next design; uncertainty and stopping decision.
Code/data source	profpark_latest_july26.png, THQ/KRICT iterative run folders where applicable, collaborator records, and final wet-lab dataset. Use

	only the chemistry actually shown in main Figure 5.
Production status	Needs final wet-lab table and identity audit

Figure S36. DPDTC protocol 1: nitrobenzoic acid-benzylamine coupling

Purpose	Provide the first replacement chemistry in revised main Figure 6 with enough detail to reproduce feed preparation and understand each refinement decision.
Recommended panels	Batch protocol; stream A/B recipes; stage topology; inventory assignment; temperature/residence-time design history; wet-lab conditions and yield; analytical trace.
Code/data source	2 protocols.pdf and canonical case_02 artifacts.
Production status	Blocked pending corrected rerun and wet-lab data

Figure S37. DPDTC protocol 2: benzoic acid-morpholine coupling

Purpose	Provide the second replacement chemistry in revised main Figure 6, emphasizing aqueous interstage addition and phase/mixing considerations.
Recommended panels	Batch protocol; stream A/B recipes; stage topology; inventory assignment; temperature/residence-time design history; wet-lab conditions and yield; analytical trace.
Code/data source	2 protocols.pdf and canonical case_03 artifacts.
Production status	Blocked pending corrected rerun and wet-lab data

Figure S38. Cross-case experimental agreement, refinement efficiency, and failure modes

Purpose	Close the ESI by summarizing what FlowPilot predicted, what was built, what was measured, and how many refinement cycles were needed across all wet-lab cases.
Recommended panels	Predicted-versus-measured yield or conversion; initial-to-final parameter shifts; cycles to target; inventory violations caught; unsuccessful screens retained; final uncertainty classification.
Code/data source	Final harmonized wet-lab table for main Figures 5 and 6.
Production status	Needs completed experiments

Supporting text sections

S1. Software architecture and versioning. Commit identifier, release date, Python environment, package versions, model endpoints, prompts, seeds where supported, retry policy, and deterministic/stochastic boundaries.

S2. Standardized intake and authority policy. Question bank, readiness requirements, unavailable responses, package freezing, and evidence/constraint authority order.

S3. Literature corpus construction. Inclusion criteria, paper deduplication, metadata extraction, manual/machine verification status, and hidden-reference exclusions.

S4. Engineering rule-base construction. Rule sources, schema, category/severity definitions, deduplication, conflict handling, and sparse-domain limitations.

S5. Deterministic equations and assumptions. All equations, units, correlations, property assumptions, thresholds, and validity ranges for the nine-step calculator.

S6. Gas-liquid and multistage conventions. Definitions of inlet/STP and in-channel flow, gas equivalents, holdup treatment, stage-flow propagation, total residence time, and quench-only operations.

S7. Inventory profile and allocation rules. Equipment schema, PDF/PPT/DOCX extraction, hard limits, accessory assumptions, quantity allocation, reactor trains, and blocked-design behavior.

S8. Council prompts and bounded authority. System prompts, specialist scopes, candidate generation, deterministic recomputation after revision, skeptic audit, and orchestrator scoring.

S9. Benchmark and reproducibility protocol. Case selection, model matrix, repeats, candidate budgets, endpoints, failed runs, exclusions, and statistical methods.

S10. Ablation protocol and interpretation. Matched-input design, feasible/infeasible controls, frozen scoring, schema-neutral analysis, bootstrap strategy, and limits on superiority claims.

S11. Experimental procedures. Full batch references, feed preparation, startup/steady-state definitions, reactor assembly, pressure/temperature control, sampling, workup, analytical methods, and safety controls.

S12. Data and code availability. Artifact manifests, raw prompts/completions, JSON schemas, logs, topology files, analysis scripts, checksums, and repository release plan.

Proposed supporting tables

Table	Title	Required content
Table S1	Software, model, and execution configuration	Exact model IDs by component, endpoint/provider, sampling controls, retry policy, embedding model, environment, and commit.
Table S2	Schema objects and field ownership	Every major field, owning module, allowed modifiers, validation rule, and final source of truth.
Table S3	Literature corpus summary	Paper and record counts by reaction class, reactor, phase, material, and verification status.
Table S4	Engineering rule-base summary	Rule counts by category, severity, source, and chemistry class.
Table S5	Equations, symbols, units, and validity ranges	Complete deterministic calculator reference.
Table S6	Model-matrix run manifest	All upstream/council pairings, repeats, completion state, latency, tokens, and artifact path.
Table S7	Candidate-budget results	Per-repeat selected design, revisions, disqualifications, pathologies, and variability.

Table S8	Ablation cases and matched inputs	Protocol family, objective, inventory state, repeat hashes, feasibility oracle, and hidden-reference exclusion.
Table S9	Ablation scoring specification	Primary endpoint, quality dimensions, weights, hard gates, and schema-neutral sensitivity.
Table S10	Per-case ablation outcomes	Qwen and GPT one-shot versus full-pipeline results without averaging away failures.
Table S11	KHU laboratory inventory	Pumps, tubing, reactors, lights, MFCs, BPRs, connectors, temperature/pressure limits, quantities, and assumptions.
Table S12	Photoredox wet-lab cycles	Complete design and measured result for every cycle, including unsuccessful and rejected conditions.
Table S13	DPDTC feed-preparation and operating conditions	Masses, concentrations, reservoir volumes, flow ratios, stage flows, residence times, temperatures, pressures, yields, and analytical method.
Table S14	Cross-case deviations and unresolved assumptions	Difference between proposed and executed design, reason, consequence, and resolution.

Publication gates before figure production

These are scientific consistency gates, not cosmetic edits

The two DPDTC outputs presently contain conflicts that must be corrected before they are shown as experimental recommendations. Figures S36-S38 and revised main Figure 6 should be produced only from a new frozen, internally consistent dataset.

- Correct DPDTC protocol 1 stream stoichiometry. The current topology uses equal A/B flow while describing B as 2.10 M benzylamine, which does not preserve 1.05 equivalents.
- Resolve the 95 °C batch condition versus the KHU inventory maximum of 80 °C. A deterministic clamp cannot be presented as chemically equivalent without a residence-time screen or verified 95 °C hardware rating.
- Regenerate final operating instructions after inventory reconciliation. Current retained text can say 95 °C/7 bar while the final contract says 80 °C/2.5 bar.
- Freeze one authoritative wet-lab data table and use it for main figures, ESI figures, captions, and JSON artifacts. Do not reconstruct values independently in multiple graphics.
- Retain failed, rejected, and low-yield cycles. Removing them would turn the iterative evaluation into a selected-success narrative.
- Use the matched Qwen and GPT architecture comparisons as the primary ablation. Label older composite-score analyses as exploratory or omit them where the metric favored no-council variants.
- Do not claim wet-lab yield superiority from the architecture ablation. The ablation supports constraint-compliant executability and repeatable design behavior.
- Confirm that Figure 5's wet-lab chemistry is the photoredox Giese/aerobic oxidation case. Do not mix THQ iterative records into that figure unless the manuscript explicitly introduces THQ as a separate chemistry.

Ablation figures to use and figures to avoid

Primary ESI evidence: the matched Qwen-one-shot versus Qwen-FlowPilot and GPT-one-shot versus GPT-FlowPilot study, the frozen endpoint/scoring specification, schema-neutral sensitivity, deployment gates, and the evidence-first validation package.

Exploratory only: the older architecture composite in which no-council or no-retrieval variants can outrank the full pipeline because formal validity, rejection behavior, and decision assurance are incompletely represented. These plots should not be used as the primary superiority evidence. They may be archived in the repository or described as development-stage diagnostics.

Recommended production order

1. Freeze the revised manuscript claims, exact model identifiers, and final list of wet-lab chemistries.
2. Correct and rerun both DPDTC cases under a coherent 80 °C screen or a documented 95 °C-capable inventory; freeze feed recipes and final-design contracts.
3. Create one master wet-lab CSV with all successful and unsuccessful cycles and generate main Figures 5-6 plus ESI Figures S34-S38 from that file.
4. Freeze the matched ablation manifest and regenerate Figures S23-S31 using manuscript-consistent labels and vector output.
5. Regenerate engineering and inventory Figures S11-S17 from the final code revision, then run arithmetic/topology audits.
6. Regenerate corpus/retrieval Figures S7-S10 from the exact dataset version cited in Methods.
7. Capture GUI/provenance Figures S3-S6 last, after the interface and artifact format are frozen.
8. Perform a final cross-document audit: every number in the main text, ESI, tables, captions, JSON, and diagrams must derive from the same source row or result contract.

Deliverable summary

Supporting figures	38 proposed figures (S1-S38)
Supporting tables	14 proposed tables (S1-S14)
Text sections	12 method/data sections
Main-figure change	Figure 5 gains wet-lab refinement; Figure 6 becomes two DPDTC wet-lab protocols
Moved figure	Current alpha-bromination Figure 6 becomes Figure S32, with diagnostics in Figure S33